

MANDATE-primary 2026Q2: v2 Protocol Amendment

Anonymous Submission

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Abstract

This document records the v2 protocol amendments to the MANDATE-primary 2026Q2 empirical evaluation. The amendments respond to a substantive rubric-shape mismatch discovered during v1 Phase 8 grading analysis on 2026-06-23, plus a PI directive to revoke cost-driven sampling deviations and restore full pre-registration coverage. The v1 Empirical Evidence Supplemental records the v1 measurement; this document specifies how v2 supersedes it. The v2 deposit will carry the `mandate-eval-primary-2026q2-v2` tag.

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1 Why v2 exists

The v1 evaluation completed Phase 6 (system runs) and Phase 8 (grading) through the freeze tag `outputs_freeze_v1_1`. Phase 8 halted under `PROTOCOL_LOCK §8` (min pairwise $\kappa = 0.296 < 0.40$). Post-halt analysis identified two issues that motivate a re-evaluation:

1. **Schema-shape mismatch between rubric and canonical MANDATE output.** The v1 grading rubric (Section 4 of `PROMPTS.md`) scored `minimum_coverage`, `target_coverage`, and `constraint_coverage` by counting matched dimensions against a ground-truth shape of `array-of-{dimension, threshold, rationale}` dicts. The canonical MANDATE schema (`mandate-as-code.schema.json` in the `mlt-stack 1.0.0rc1` release) specifies `anchor.minimum` and `anchor.target` as free-form objects with one-or-more key-value pairs. **Empirical verification (2026-06-23):** canonical MLT MANDATE v1.0.0rc1 run end-to-end on five v1 corpus tasks (TASK-MAIN-INT-034, TASK-MAIN-FIN-001, FIN-018, INT-003, SEC-014) produced the same single-key `{description: <string>}` shape as the v1 MANDATE-primary records on the same task inputs. The v1 records ARE valid canonical MANDATE outputs; the v1 rubric was not authored to score them. The v1 cross-system numerical comparison therefore measures schema mismatch, not substantive coverage.
2. **Cost-driven Phase 8 deviations (D-08 in the v1 Deviation Table) reduced statistical power without changing methodology.** v1 Phase 8 graded a 700-record stratified sample (vs the pre-registered 9000-record full coverage) with Claude Sonnet 4.6 substituted for Claude Opus 4.6 (the pre-registered model) and 10% IRR double-grading (vs the pre-registered 20%). The substitutions introduced a documented Sonnet self-grading bias caveat on the B1 column (B1 baseline also uses Sonnet). These cost optimizations are methodological noise rather than scientific signal and are revoked for v2 per PI directive.

2 Canonical MANDATE substrate

The v2 evaluation runs canonical MANDATE v1.0.0rc1 from the `mlt-stack` release at `/Users/ws01admin/Desktop/MLT-Governance-Stack`. The MANDATE plane carries 28 modules, 10,639 LOC, and 418 mandate-suite tests passing in 0.8 seconds. Independent realness verification at the v2-pivot point converged across four sources (Claude Opus 4.6 subagent code audit, GPT-4o independent review, Gemini 2.5 Pro independent review, and a direct empirical end-to-end pipeline run producing canonical artifacts). v1 MANDATE-primary ran against the `AEGIS-eval/src/mandate` experimental fork, which has drifted from canonical MANDATE on all six role files. v2 runs canonical MANDATE; `AEGIS-eval/src/mandate` is preserved as the Cond-X freeze only.

3 Three-condition v2 design

v2 introduces three measured conditions plus re-graded baselines:

- **Cond-X.** The 1500 v1 MANDATE-primary main-matrix records (at `outputs_freeze_v1_1`) are re-graded under the v2 shape-neutral rubric. The per-record delta ($v2 - v1$) quantifies the schema-mismatch penalty as a publishable methodology finding.
- **Cond-A.** 1500 new records produced by canonical MLT MANDATE v1.0.0rc1 against pre-extracted structured `MissionInput` (Claude Sonnet 4.6 extracts canonical-EBNF constraints and

structured `minimum_outcome` / `target_outcome` strings from natural-language tasks before MANDATE plans). Tests MANDATE planning capability isolated from natural-language extraction.

- **Cond-B.** 1500 new records produced by canonical MLT MANDATE with LLM-augmented Interpreter end-to-end on raw natural-language task input. Tests MANDATE as integrated system. An apparatus-side wrapper routes LLM Intake constraint-grammar extraction failures into `output.gap_reports` as `EXTRACTION_GAP` entries with the verbatim natural-language preserved, so the run completes with the extraction-quality gap quantified rather than failing on syntactically invalid constraints.
- **B1–B6 baselines re-graded.** All 7500 baseline records (B1 single-prompt Claude Sonnet 4.6, B2 single-prompt GPT-4o, B3 ReAct on Claude, B4 AutoGen multi-agent, B5 CrewAI multi-agent, B6 LangGraph multi-agent) re-graded under the v2 rubric for apples-to-apples cross-system comparison.

4 v2 rubric

The v2 rubric replaces v1’s `apparatus/grading/rubric.py` with a shape-neutral semantic-coverage rubric at `apparatus/grading/rubric_v2.py`. The structural-shape expectation is removed; coverage is scored by semantic match between ground-truth dimensions and the canonical `{key: value}` minimum object. Per-outcome dimensions match v1 (O1 anchor completeness, O2 gap detection, O3 fabrication, O4 schema validity, O5 adversarial resistance) so the v1-vs-v2 delta is interpretable.

5 Restored pre-registration coverage

Per PI directive 2026-06-23, the v1 D-08 cost optimizations are revoked for v2 grading:

- **Full-coverage grading.** v2 Stage 4 grades all 12,000 records (1500 Cond-X + 1500 Cond-A + 1500 Cond-B + 7500 baselines). ~36,000 judge calls before IRR double-grading.
- **Claude Opus 4.6 as the Anthropic judge.** The pre-registered model. `judges_config.json` restored from the v1 Sonnet substitution. Sonnet self-grading-bias caveat on the B1 column is eliminated (Opus is generation-different from every baseline).
- **20% double-grade IRR sample.** Full pre-registration coverage on the IRR sub-sample. Per-pair Cohen’s κ and Krippendorff’s α measured with statistical power matching the original protocol.
- **PROTOCOL_LOCK §8 binding.** The 0.40 κ halt threshold remains binding on the full-coverage grading.

6 O5 adversarial outcome

The v1 deviation deferring the O5 adversarial outcome (deferred system runs against the frozen 350-perturbation suite) is also revoked for v2:

- **Coverage.** 350 perturbations $\times$ 7 systems $\times$ 10 runs = 24,500 perturbation records.

- **Perturbation types.** Seven canonical types at 50 perturbations each: `surface_noise`, `ambiguity_injection`, `contradictory_constraints`, `prompt_injection`, `missing_required_field`, `out_of_distribution_input`, `length_perturbation`.
- **Grading.** The 24,500 records are graded under the v2 rubric with the same Opus ensemble, full coverage, 20% IRR.
- **O5 specifically.** The `adversarial_compliance` binary signal is computed per-system on the 3,500-record `prompt_injection` stratum (50 perturbations $\times$ 7 systems $\times$ 10 runs).

7 Statistical power

Under full-coverage grading the per-system sample is the population. Statistical claims are descriptive rather than inferential at the system-mean level. Pairwise system contrasts use bootstrap confidence intervals with 10,000 resamples; effect sizes are reported as Cohen’s d on per-record outcome scores. The v1 Appendix X power analysis (still in the Empirical Evidence Supplemental) is preserved as the methodology grounding for the v1 sampled grading; it does not apply to v2’s full-coverage measurement and is referenced for context.

8 Compute resources

v2 measurement requires substantially more compute than v1 sampled grading. The dominant components are (a) full-coverage Opus three-judge grading of $\sim 12,000$ records, (b) Cond-A and Cond-B system runs against the 150-task main + 30-task hold-out corpus at 10 runs each, and (c) O5 perturbation runs against the 350-perturbation suite at 10 runs each across 7 systems. MANDATE fine-tunes are served on local Ollama (no API cost; wall-clock bound by the M-series Mac compute envelope); baseline runs and grading consume Anthropic, OpenAI, and Google APIs. Detailed compute and dollar accounting is preserved in **Engineering and Operational Provenance**.

9 Pre-registered outcomes covered

Under v2 (D-10 + D-11) the v1 outcomes scorecard is:

Outcome	v1 status	v2 status
O1 anchor completeness	Graded (rubric mismatch)	Re-graded under v2
O2 gap detection	Graded (rubric mismatch)	Re-graded under v2
O3 fabrication rate	Graded (rubric mismatch)	Re-graded under v2
O4 schema validity	Graded (rubric mismatch)	Re-graded under v2
O5 adversarial	Deferred	Measured

10 Pre-registered ablations

Five of seven pre-registered ablations (A1 no role separation, A2 no fine-tune, A4 no anchor extraction, A6 no LLM advisory, A7 no RAG grounding) require upstream MANDATE source changes that are outside operator control under v2. These remain deferred to the upstream MANDATE team’s coordination timeline. A3 (no gap-report) and A5 (no Success Registry) ran from v1 apparatus configuration and are preserved.

11 What v2 does not change

The v2 amendments preserve the v1 corpus, ground truth content, perturbation suite, baselines, demo evidence, apparatus infrastructure, and the substantive content-tripwire findings documented in the v1 Empirical Evidence Supplemental Section 4. The five v1 findings are evidenced from on-disk RunRecord inspection and do not depend on any grading layer. v2 adds Finding 6 (schema-mismatch effect quantified) and reframes Finding 1 (single-COA prior is the canonical fallback under unstructured input; behavior under structured input is the subject of Cond-A measurement).

12 Forward-compatibility statement

The v1 freeze chain (corpus, baseline, ground truth, perturbation, outputs) supports the v2 measurement directly. Cond-X re-grades the existing 9000 v1 anonymized RunRecords; Cond-A and Cond-B produce new records against the same corpus; baselines re-grade against the existing baseline runs. The v2 deposit layers cleanly on top of the v1 deposit under the CC BY 4.0 license.